

Acoustic-Based Detection, Classification, and Localization of Impulsive Sounds

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⟨CONFERENCE NAME — TO BE CONFIRMED⟩

September 16, 2026

[0:00–0:30]

Good morning. // My name is Martin Maxa, from the Czech Technical University in Prague, Department of Measurement. // I am presenting joint work with Doctor Jakub Svatoš. //

Our topic is acoustic detection, classification and localization of impulsive sounds — in practice, gunshots. // I will show you how we detect them, how we tell them apart, and how we find where they came from. //

(// = pause. [CLICK] = advance the slide.)

Why acoustic surveillance

- Shootings in public spaces are more frequent; global gun ownership is rising.
- Growing demand for autonomous security in communal areas — campuses, hospitals, parks, government facilities.
- Cameras, drones and security personnel give *some* protection, but fall short in high-risk situations with an active shooter.

What an acoustic system adds

Identifies gunfire, pinpoints its source, and tracks an ongoing threat — faster detection and shorter response times for security forces.

Acoustic Detection of Impulsive Sounds

└ Motivation

└ Why acoustic surveillance

Why acoustic surveillance

- Shootings in public spaces are more frequent; global gun ownership is rising.
- Growing demand for autonomous security in communal areas — campuses, hospitals, parks, government facilities.
- Cameras, drones and security personnel give some protection, but fall short in high-risk situations with an active shooter.

What an acoustic system adds

Identifies gunfire, pinpoints its source, and tracks a non-engaged threat — faster detection and shorter response times for security forces.

[0:30–1:15]

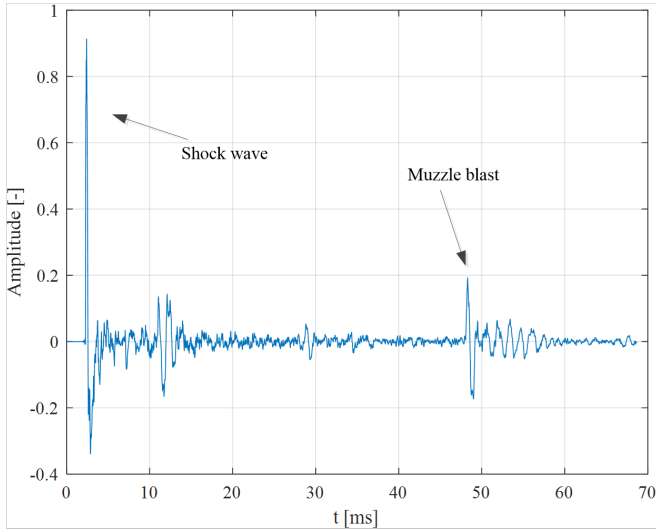
Shootings in public spaces are becoming more frequent, and gun ownership is rising worldwide.

// At the same time there is growing demand for autonomous security in places where people gather — campuses, hospitals, parks, government buildings. //

Cameras, drones and security staff already give some protection. But in a high-risk situation, with an active shooter, they fall short. // A camera has to be pointed the right way. A guard has to be in the right place. //

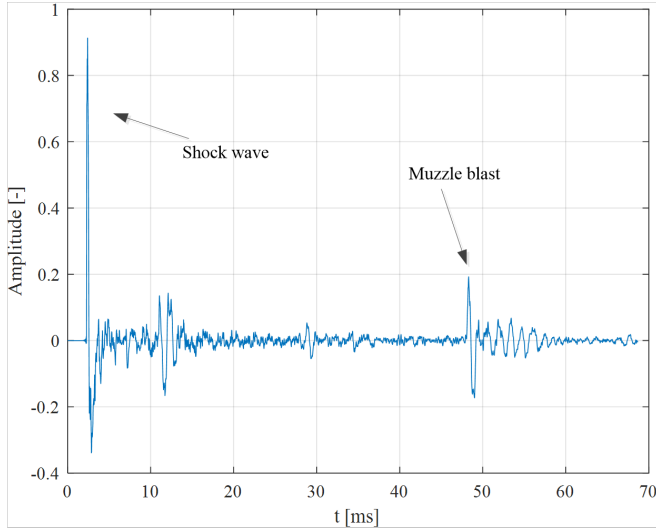
Sound does not care about line of sight. An acoustic system hears the shot and tells you where it came from. //

The acoustic signature of a gunshot



- **Shock wave** (“N-wave”) — arrives *first*; exists only when the bullet is supersonic ($v > c$).

The acoustic signature of a gunshot



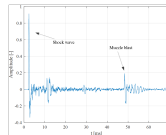
- **Shock wave** (“N-wave”) — arrives *first*; exists only when the bullet is supersonic ($v > c$).
- **Muzzle blast** — gases leaving the barrel, 1–3 ms; travels at the speed of sound, so it arrives ≈ 45 ms later.
- Everything between and after: **reflections**.

Acoustic Detection of Impulsive Sounds

└ Signal

└ The acoustic signature of a gunshot

The acoustic signature of a gunshot



- Shock wave ("Boum") — arrives first, usually when the bullet is supersonic ($v > c$).
- Muzzle blast — gases leaving the barrel, 1-2 ms traveling at the speed of sound, so it arrives next ms later.
- Everything between and after: reflections.

[1:15–2:30]

So what does a gunshot look like to a microphone? // This is a real recording. Two components define the pattern — and notice that they do not arrive together. //

The first spike, at about two and a half milliseconds, is the shock wave. It exists only when the bullet travels faster than sound. The bullet drags a cone of compressed air behind it, and that cone sweeps across the microphone. That is the characteristic N shape. //

[CLICK]

The second spike, forty-five milliseconds later, is the muzzle blast — the explosive gases leaving the barrel. It leaves the weapon at the instant of firing, but it only travels at the speed of sound, so it arrives late. It lasts one to three milliseconds. //

Now look at everything in between and after. That is no longer the gunshot — those are reflections from the surroundings. Remember that; it comes back at the end of the talk. //

Why naive processing is not enough

- **Spectral analysis alone** often captures environmental noise rather than the gunshot itself.
- **Time-domain patterns** get distorted by nonlinear dispersion and obstacles.

Consequence

Effective feature extraction must integrate *both* time and frequency domain processing.

Acoustic Detection of Impulsive Sounds

└ Signal

└ Why naive processing is not enough

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[2:30–3:00]

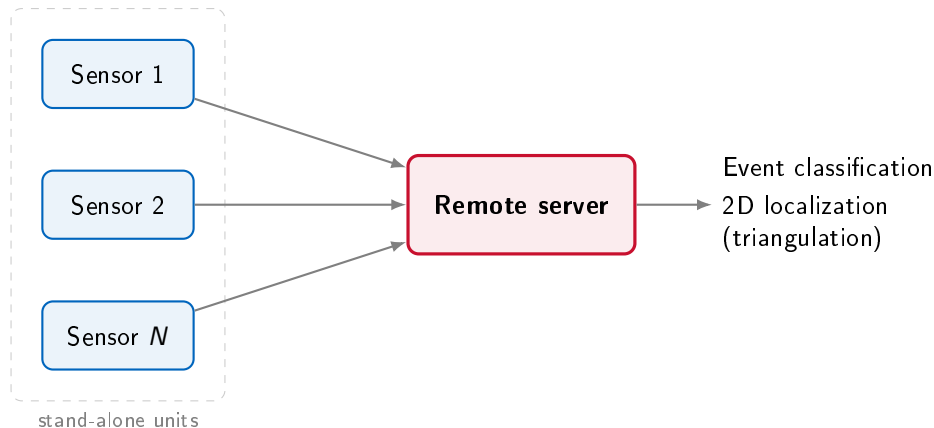
You might think this is easy — just look at the spectrum, or just look at the waveform. //

Neither works on its own. //

Spectral analysis alone tends to describe the environment rather than the gunshot. // And the time-domain pattern gets distorted by nonlinear dispersion and by obstacles in the path. //

So feature extraction has to combine both domains. That is what the rest of the talk is about. //

Proposed system architecture



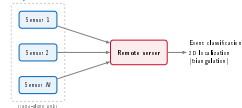
- Each sensor continuously monitors its surroundings and transmits detected signals to the server for advanced processing.
- Multiple units give broader coverage and adaptability to environments.
- Tested with various firearm calibers and ammunition.

Acoustic Detection of Impulsive Sounds

└ System

└ Proposed system architecture

Proposed system architecture



- Each sensor continuously monitors its surroundings and transmits detected signals to the server for advanced processing.
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[3:00–4:00]

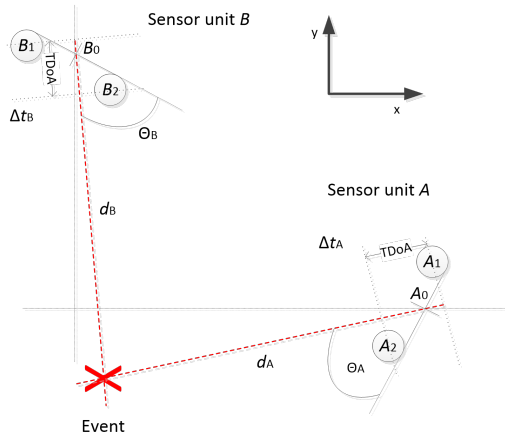
Here is the system. // Each sensor is a stand-alone unit. It listens continuously, and when it detects something impulsive, it sends that signal to a remote server. //

The server does two things. It classifies the event — is this a gunshot, and what kind. And it determines where the event happened, in two dimensions. //

Why multiple units? Two reasons. Coverage — one unit hears one neighbourhood. And localization — you need at least two bearings to get a position, and I will show you that on the next slide. //

The system has been tested with several firearm calibers and several types of ammunition. //

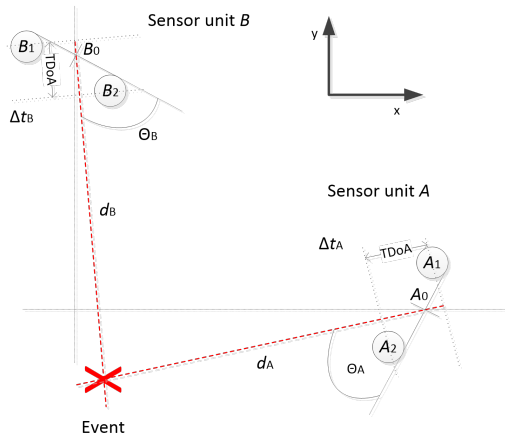
Locating the source



$$\Theta = \arcsin\left(\frac{c \Delta n}{d f_s}\right)$$

- Two MEMS microphones per unit ($d = 0.186$ m, $f_s = 44.1$ kHz) give one **bearing** Θ .

Locating the source



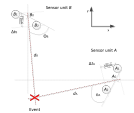
$$\Theta = \arcsin\left(\frac{c \Delta n}{d f_s}\right)$$

- Two MEMS microphones per unit ($d = 0.186$ m, $f_s = 44.1$ kHz) give one **bearing** Θ .
- Two units, two bearings — they **intersect** at the event.
- Measured: cross-correlation with parabolic interpolation, **MAE** 0.77° over 70 events (-45° to $+60^\circ$).

Acoustic Detection of Impulsive Sounds

└ System

└ Locating the source



$$\Theta = \arcsin\left(\frac{c \Delta t}{d}\right)$$

- Two MEMS microphones per unit
[$d = 0.188\text{ m}$, $f_s = 44.1\text{ kHz}$] give us bearing Θ .
- Two units, one bearing — they intersect at the source.
- Measured: cross-correlation with parabolic interpolation, MAE 0.27° over 78 events in $[-45^\circ$ to $+88^\circ]$.

[4:00–5:00]

So how do we actually find the shooter? //

Each sensor unit carries two MEMS microphones, eighteen point six centimetres apart. // The sound reaches one microphone before the other. That delay is tiny — at most twenty-four samples at forty-four kilohertz — but it is enough. //

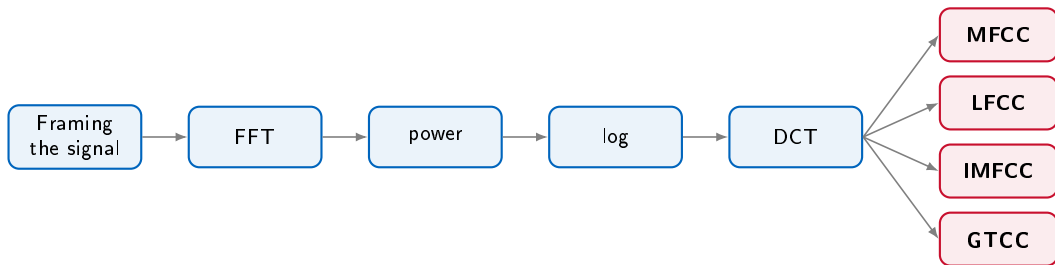
From it we compute the bearing to the source, theta: the speed of sound times the delay in samples, divided by the microphone spacing times the sampling frequency. //

[CLICK]

One unit gives you a direction. Two units give you two directions, and where they cross is the position. //

We measured how good that bearing is. Using cross-correlation with parabolic interpolation, over seventy recorded events, the mean absolute error is zero point seven seven degrees. //

Feature extraction — cepstral coefficients



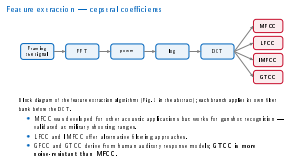
Block diagram of the feature extraction algorithms (Fig. 1 in the abstract); each branch applies its own filter bank before the DCT.

- MFCC was developed for other acoustic applications but works for gunshot recognition — validated at military shooting ranges.
- LFCC and IMFCC offer alternative filtering approaches.
- GFCC and GTCC derive from human auditory response models; **GTCC is more noise-resistant than MFCC.**

Acoustic Detection of Impulsive Sounds

└ Features

└ Feature extraction — cepstral coefficients

**[5:00–6:15]**

Now the classification. // The feature extractor is based on cepstral coefficients. The chain is the same for all of them: cut the signal into frames, take the FFT, take the power, take the logarithm, and finish with a discrete cosine transform. //

What differs is one step that is not drawn here — the filter bank that sits before the DCT. //

Mel-frequency coefficients, MFCC, were developed for speech, but they work for gunshots; that has been validated at military shooting ranges. // LFCC uses linear spacing, IMFCC an inverted mel scale. // And GTCC uses a gammatone filter bank, which models the response of the human ear. It is known to be more noise-resistant than MFCC. //

We tested all four, at three frame lengths — fifteen, thirty and fifty milliseconds — with two classifiers: a neural network and a support vector machine. //

Classification results — 9 mm class

Frame	Extractor	ACC (%)		RLC (%)		MCC (–)	
		SVM	NN	SVM	NN	SVM	NN
15 ms	MFCC	83.90	85.59	80.65	67.74	0.62	0.62
	LFCC	83.90	83.90	80.65	64.52	0.62	0.57
	IMFCC	85.59	90.68	83.87	87.10	0.66	0.77
	GTCC	93.22	97.46	87.10	93.55	0.82	0.93
30 ms	MFCC	84.75	87.29	67.74	87.10	0.60	0.70
	LFCC	87.29	83.90	70.97	70.97	0.66	0.59
	IMFCC	85.59	92.37	67.74	80.65	0.62	0.80
	GTCC	97.46	98.31	90.32	93.55	0.93	0.96
50 ms	MFCC	84.75	93.22	74.19	90.32	0.61	0.83
	LFCC	88.98	86.44	77.42	77.42	0.71	0.66
	IMFCC	89.83	90.68	74.19	80.65	0.73	0.76
	GTCC	97.46	99.15	90.32	96.77	0.93	0.98

Table I in the abstract — example of classification results for a 9 mm class. Test partition: 118 events, 31 of them 9 mm.

Acoustic Detection of Impulsive Sounds

└ Results

└ Classification results — 9 mm class

Classification results — 9 mm class

Frame	Extractor	ACC (%)		REC (%)		MCC (+)	
		SVM	NN	SVM	NN	SVM	NN
15 ms	MFCC	85.80	85.0	88.88	87.74	0.82	0.82
	LFCC	85.88	85.8	88.85	84.53	0.82	0.80
	IMFCC	85.58	85.8	88.87	87.28	0.80	0.77
	GTCC	85.32	87.48	87.18	83.55	0.82	0.83
30 ms	MFCC	84.75	87.28	87.74	87.28	0.80	0.79
	LFCC	87.28	85.8	88.87	86.87	0.80	0.80
	IMFCC	85.58	87.7	87.74	88.85	0.82	0.83
	GTCC	87.48	88.28	88.22	83.55	0.83	0.80
50 ms	MFCC	84.75	88.22	84.18	88.22	0.80	0.83
	LFCC	88.88	88.4	87.43	87.43	0.75	0.80
	IMFCC	88.83	88.8	84.18	88.85	0.73	0.79
	GTCC	87.48	88.25	88.22	88.77	0.83	0.80

Table in the abstract — example of classification results for a 9 mm shot. The positions (10 shots, 20 of them 9 mm).

[6:15–7:45]

This is the result table for the nine-millimetre class. Let me walk you through it. //

The rows are grouped by frame length — fifteen, thirty, fifty milliseconds — and inside each group you see the four extractors. // The columns are three metrics, each for both classifiers: accuracy, recall, and the Matthews correlation coefficient. //

Now look only at the red rows. That is GTCC. // At every single frame length, and for both classifiers, GTCC is the best row in its block. //

The best cell in the table is GTCC with the neural network at fifty milliseconds: ninety-nine point one five percent accuracy, ninety-six point seven seven percent recall, and a Matthews coefficient of zero point nine eight. //

To put those numbers in context — the test set has one hundred and eighteen events, thirty-one of them nine-millimetre. So that accuracy is one hundred and seventeen correct out of one hundred and eighteen, and that recall is thirty shots out of thirty-one. //

What the numbers say

- MFCC, LFCC and IMFCC yield similar results.
- **GTCC consistently performs better** across every frame length and both classifiers.
- **NN outperforms SVM** — better at modelling complex patterns. The gap widens with longer frames.
- **Longer frames improve accuracy**, but start to capture environmental reflections, which can pull focus away from the gunshot itself.

Best configuration in the table

GTCC + NN at 50 ms — 99.15 % ACC, 96.77 % RLC, MCC 0.98.

Acoustic Detection of Impulsive Sounds

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Best configuration in the table

GTCC + NN at 50 ms — 99.35 % ACC, 99.77 % ROC, MCC 0.99.

[7:45–8:45]

Three things come out of this. //

First, MFCC, LFCC and IMFCC behave almost identically. Changing the shape of a mel-type filter bank does not buy you much. //

Second, GTCC is consistently better — every frame length, both classifiers. The gammatone filter bank resolves the low frequencies more finely, and that is exactly where the energy of a muzzle blast sits. //

Third, the neural network beats the support vector machine, and the gap grows with longer frames. //

And about those longer frames — they do improve accuracy, but remember the reflections I showed you at the beginning. A fifty-millisecond frame starts to include the acoustics of the room instead of the gunshot. So this trend does not continue forever. //

Conclusion

- A distributed network of stand-alone acoustic sensors with a remote server detects, classifies and localizes impulsive sounds in 2D.
- Cepstral coefficients are an effective feature extractor for gunshot classification with both NN and SVM.
- GTCC beats Mel-frequency based coefficients; NN achieves slightly better classification success, especially with longer frames.
- Validated with various firearm calibers and ammunition.

Thank you — questions?

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Acoustic Detection of Impulsive Sounds

└ Conclusion

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Thank you — questions?

source@falconer.cc manam@falconer.cc

[8:45–9:30]

To summarise. //

We built a distributed network of stand-alone acoustic sensors with a remote server, which detects, classifies and localizes impulsive sounds in two dimensions. //

Cepstral coefficients work well as a feature extractor for gunshot classification, with both classifiers. // Gammatone-based coefficients beat the mel-frequency family, and the neural network is slightly better than the SVM, especially at longer frames. //

And all of this was validated with real firearms, several calibers and several types of ammunition. //

Thank you. I am happy to take questions. //

Backup — experimental setup and dataset

- Military shooting range, fixed sensor position, clear line of sight to every firing position.
- Four weapons: 9 mm, 5.56 NATO SD, 7.62 mm Tokarev, .22 — each with a single ammunition type.
- Plus a false-alarm class with gunshot-like impulsive patterns.
- 26 features per extractor; $\approx 75 / 25$ train–test split.
- NN: MATLAB Neural Fitting, Levenberg–Marquardt, 370 iterations.

Class		Train	Test
0	False alarms	86	21
1	9 mm	70	31
2	5.56 NATO SD	100	31
3	7.62 mm Tokarev	55	17
4	.22	60	18
Total		371	118

Svatoš & Holub, *Meas. Sci. Technol.* **35** (2024)
076122, Table 2.

Backup — bearing estimation methods

Method	MAE	Median	Max
Cross-correlation	0.84°	0.41°	7.61°
Cross-correlation + parabolic interp.	0.77°	0.56°	6.67°
Phase-based delay estimation (PBDE)	3.24°	1.80°	29.0°
PBDE + linear regression	3.91°	3.55°	12.15°

70 events, 9 reference angles from -45° to $+60^\circ$, $d = 0.186$ m, $f_s = 44.1$ kHz.

PBDE algorithm

1. FFT of both channels
2. select the relevant frequency bins
3. phase angle $\varphi = \arg(S_1 S_2^*)$
4. delay from the phase slope

How many units for 2D?

- ToA, 1 mic per unit — 3 units
- TDoA, 1 mic per unit — 4 units
- **TDoA, 2 mics per unit — 2 units**

Backup — other cepstral coefficients

- **LPCC** (Linear Prediction Cepstral Coefficients) and **PLPC** (Perceptual Linear Prediction Coefficients) — noise elimination, speech recognition.
- **GFCC** (Greenwood Function Cepstral Coefficients) — derived from human auditory response models, used for classifying animal sounds.
- **GTCC** (Gammatone-Frequency Cepstral Coefficients) — same family, greater resistance to noise than MFCC.

Backup — metrics

ACC Accuracy — share of correctly classified events.

RLC Recall, also known as *sensitivity* — share of actual gunshots of the class that were detected.

MCC Matthews Correlation Coefficient — balanced measure robust to class imbalance; +1 perfect, 0 random.

Definitions as used in the source study: Svatoš & Holub, *Meas. Sci. Technol.* **35** (2024) 076122, eqs. (1)–(3).